

Shubham Goel

Senior Machine Learning Engineer · Multimodal · RecSys · Computer Vision

217-979-1769 · shubhamgoel27@gmail.com · shubham.gg · github.com/shubhamgoel27 · linkedin.com/in/shubhamgoel27

EXPERIENCE

Meta, Instagram Ads Ranking

2024 – 2026

Senior Machine Learning Engineer, Menlo Park

- Built Instagram Ads' **Video OCR pipeline**: quality-ranked frame sampling, SSIM-based deduplication, and a **domain-tuned OCR backbone** within 1.5pp F1 of the production VLM (Llama 4 Scout) at 80× lower inference cost.
- Shipped it to production: **0.06–0.08% online NE gain** (normalized entropy, the ranking model's loss), 0.12% on Reels, **\$30M+ annual revenue lift**; basis of a first-author RecSys '26 paper.
- Led a **cross-app data augmentation** effort integrating Facebook ad-click signal into the IG Ads ranking model via auxiliary learning tasks; shipped with a **0.04% NE gain** (\$20M+ annual revenue lift).
- Post-trained **Qwen3 14B** for ad understanding by distilling Claude Opus 4.7 teacher labels via SFT (+9pp extraction accuracy over base); shipped ad-side semantic ranking features (ad summary, persuasion technique, target audience) for a **0.05% NE gain**, selecting prompts, decoding parameters, and checkpoints against a per-field eval harness.
- Created **FeatureLens**: semantic search, near-duplicate detection, and cross-model gap analysis over the ads feature catalog; adopted by **80 ranking engineers**, powering ~250 new feature launches.

ZEFR

2021 – 2024

Senior Machine Learning Scientist, Remote

- Architected a **multimodal semantic search** platform on Qdrant over **1B+ image & text embeddings**, replacing keyword search for a 14-person labeling team and powering training-data sourcing for CV/NLP fine-tunes across 12 brand-safety taxonomies.
- Fine-tuned **CLIP / ViT** models on ZEFR's content taxonomy for ad image classification; scaled sparse-class annotation in production with a vector-search-based **active learning** system.
- Served fine-tuned ViT/LLM workloads at >1M images/day via a platform-agnostic ONNX pipeline; designed the upstream training stack (auto-reports, quantization, model registry) and a Metaflow embedding pipeline over 100M+ videos.
- Set the team's ML roadmap; ran senior MLE hiring loops and mentored junior engineers.

University of Illinois at Urbana-Champaign

2020

Research Assistant (during M.S.)

- Led cyclone-damage prediction combining pre/post-disaster geospatial imagery with SBA parcel insurance data; Siamese change-detection network outperformed all baselines at IoU 0.804.

Reliance Jio AI Centre of Excellence

2018 – 2019

Machine Learning Engineer, Computer Vision

- Shipped a building-detection module for dense, cluttered geospatial imagery using Reverse Mask R-CNN, improving mIoU 9.3% over the prior state of the art; built an imagery-acquisition tool that 10×'d throughput and replaced the external vendor tool across a 1,400-member GIS org.

Pitney Bowes

2016 – 2018

Software Engineer, Machine Learning

- Delivered building-footprint segmentation (IoU 0.845) and rooftop-type classification (96.5%) in MXNet; implemented PSPNet in Caffe with a custom SyncBatchNorm layer for multi-GPU training; first DL hire on the team.

TALKS & PUBLICATIONS

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| 2026 | RecSys, Industry Track (first author). “The Text on the Creative: An Under-Exploited Ranking Modality for Short-Form Video Ads.” |
| 2023 | Invited talk, Snap ML group. “Labelling Sparse Data at Scale using Vector Search.” |
| 2020 | Invited talk, GeoPython. Cyclone damage prediction with Siamese change-detection networks. |

INDEPENDENT WORK

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| 2026 | curbcheck. VLM benchmark and post-training project on real-world parking-sign understanding (SF street signs), a visual reasoning task frontier models still fail; eval harness plus fine-tuned open-weights baselines. |
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EDUCATION

University of Illinois at Urbana-Champaign

2019 – 2020

M.S. in Statistics

Cluster Innovation Centre, Delhi University

2012 – 2016

B.Tech. in Mathematics and IT

SKILLS

ML / DEEP LEARNING	Python, SQL, PyTorch, CUDA, ONNX, OpenCV
LLMs & MULTIMODAL	Multimodal LLMs (text/image/video), SFT / LoRA fine-tuning, teacher-student distillation, LLM-as-judge evals, RQ-VAE / vector quantization, CLIP / ViT
RANKING & RECSYS	Recommendation systems, CTR prediction, online A/B experimentation, multi-task & auxiliary learning, candidate retrieval, embeddings
ML INFRA & RETRIEVAL	Distributed inference, Qdrant, HNSW, Metaflow, Docker, AWS Batch, Weights & Biases